# Escape, Cost, and Correlation at the Verification Floor
## of Gated Agentic Computation

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**Abstract**

Two companion papers establish a *verification floor* for gated agentic systems: *A Fault-Tolerance Threshold for Gated Agentic Computation* [2] shows that a gate family's shared blind spot—a stealth mass $\lambda_{\rm st}$, decomposing as $\lambda_{\rm st} = \lambda_{\rm red} + \lambda_{\rm irr}$—caps the reliable horizon at $H_{\rm gated} \approx H_{\rm raw}/\lambda_{\rm st}$ regardless of gate count; and *Generated Gates Inherit Their Generator's Blind Spots* [1] shows that a system generating its own gates cannot fall below that floor, which lowers only as *exogenous* model families or *reality-grounded* gates are imported. Taking that floor as given, this paper develops three consequences at the floor and draws one practical framing; the underlying identities and inequalities are classical and attributed to their sources. **(i) A series-system escape identity.** Modelling an output as $n$ units each carrying an undetected fault with (fixed) probability $f$—the accept-conditioned per-unit escape probability—the chance the output contains at least one escaped fault is $E(n) = 1 - (1 - f)^n \to 1$ for any fixed $f > 0$; the size-axis reading of Part I's horizon ceiling, with the same stealth-mass floor on $f$. **(ii) A conditional cost comparison.** Under an assumed power-law error–compute relation with exponent $p$ and a per-family stealth multiplier, reaching a target escape rate a distance $\delta$ above the shared floor costs $O(\log(1/\delta))$ by gating versus $O((1/\delta)^{1/p})$ by scaling the executor's own error down—a model calculation that formalizes the currently empirical "verify rather than scale" claim. **(iii) A correlated-escape bound.** When output units are monotone functions of shared independent upstream faults, escape events are positively associated, so $E_{\rm corr}(n) \le 1 - (1 - f)^n$ for any number of shared parents—an instance of the classical association / series-system bound, with a mutual-independence equality characterization. Finally, the floor explains a proposed ordinal assurance ladder (why internal rigor cannot substitute for independent validation) and a typed-evidence non-collapse rule. We give the epistemic/diagnosability reading of the floor that ties these to established theory, state each result's assumptions and honest caveats, and flag open questions.

## 1 Introduction

Part I [2] asked how far *verification* can extend the reliable horizon of a fixed, unreliable executor, and answered with three results in a standard constant-hazard model: a logarithmic-overhead threshold theorem, a $1/\lambda_{\rm st}$ horizon ceiling set by the verification stack's shared blind spot, and a Young–Daly cost-optimal checkpoint interval. The headline of that work is that the binding constraint on autonomous horizon is not raw model capability but the *diversity* of the verification stack.

This paper takes that verification floor—established statistically in [2] and mechanistically in the companion [1]—as given, and develops three of its consequences plus a practical framing. None is claimed as new mathematics; the identities and inequalities are classical, attributed in place and in §9. What is ours is the synthesis and, in §5, one closed-form cost comparison we did not find stated elsewhere. Our contributions:

- **A size-axis reading of the horizon ceiling (§3).** The standard series-system escape identity $E(n) = 1-(1-f)^n$, read as the output-size analogue of Part I's time-axis ceiling, making precise that "zero faults at scale" requires driving the per-unit floor to zero, not adding verification.
- **A conditional cost comparison (§5).** A model calculation comparing the cost of reaching a target escape rate by gating versus by scaling the executor's own error, with the shared floor made explicit; it formalizes a currently empirical folk claim.
- **A correlated-escape bound (§6).** The classical association / series-system bound applied to output provenance: shared provenance, at fixed unit marginals, can only lower the probability of at least one escape relative to independence, for any number of shared parents, with a mutual-independence equality characterization.
- **An assurance framing the floor grounds (§7).** A proposed ordinal evidence ladder and a typed-evidence non-collapse rule—both light repackagings of assurance-case practice—which the floor explains: internal rigor is bounded by it, and heterogeneous evidence carries structure a scalar destroys.

Along the way, §4 records an epistemic/diagnosability reading of the floor that connects it to established theory (epistemic logic, rough sets, discrete-event diagnosability). We recap Part I's notation (§2), develop the three results (§§3–6), draw the assurance framing (§7), and close with open problems (§8) and related work (§9).

## 2 The model, recapped

We adopt the model and notation of Part I [2] and restate it only to fix terms; the reader is referred there for the three theorems and their proofs.

Agentic work proceeds at a constant fault hazard $\lambda$ per unit of work and is divided into checkpointed *segments* of length $s$; a segment is *bad* with probability $p = 1 - e^{-\lambda s}$. At each checkpoint a stack of $m$ imperfect *gates*, combined by majority vote, decides whether to accept or recompute the segment; each gate has a false-accept rate $\alpha$ and a false-reject rate $\beta$ (T1 below assumes both $< \frac{1}{2}$), and $\bar{\alpha}_m, \bar{\beta}_m$ denote the effective majority-vote rates of the stack. Because a rejected segment is recomputed, the emitted segment's error rate is conditioned on acceptance—Part I's per-segment slip probability $\varepsilon_{\text{seg}}$, which we reuse in §3. A *gate family* is a set of gates sharing a mode of judgement. The family's *stealth mass* $\lambda_{\text{st}}$ is the fraction of bad artifacts that the entire family accepts—its shared blind spot; it decomposes as $\lambda_{\text{st}} = \lambda_{\text{irr}} + (1-\lambda_{\text{irr}})\rho$ into a specification/intent-bound *irreducible* floor $\lambda_{\text{irr}}$ (faults that satisfy the checkable contract but violate intent) and a lineage-correlated *reducible* part governed by a per-family probability $\rho$, with $k$ independent families driving the reducible part to $\rho^k \to 0$ but never touching $\lambda_{\text{irr}}$.

Part I established, in this model: (T1) a threshold theorem—*when the stealth mass is zero* ($\lambda_{\text{st}} = 0$) and $\max(\alpha, \beta) < \frac{1}{2}$, a stack of $m = O(\log(T/(\varepsilon s)))$ gates per checkpoint achieves end-to-end success $\geq 1-\varepsilon$ over horizon $T$ at multiplicative overhead $O(\log T)$; (T2) a horizon ceiling—when $\lambda_{\text{st}} > 0$, $H_{\text{gated}} \leq H_{\text{raw}}/\lambda_{\text{st}}$ for every $m$, and $H_{\text{gated}} \leq H_{\text{raw}}/\lambda_{\text{irr}}$ for any diversity breadth; and (T3) a cost-optimal checkpoint interval $s^\star \approx \sqrt{G/\lambda}$ with $G = mg$ the per-checkpoint verification cost. We build on (T1)–(T3) throughout and do not reprove them.

## 3 The series-system escape identity

Part I bounds reliability along the *time* axis: an output produced over a horizon, verified segment by segment. Many outputs are better described by their *size*: a document of $n$ claims, a program of $n$ functions, a data record of $n$ fields. We give the size-axis analogue and show it subsumes the horizon ceiling.

We first fix the sampling process, since the right per-unit quantity depends on it. A candidate unit is generated and passed to the verification gate; on rejection it is recomputed and re-checked until a candidate is accepted, and that accepted candidate is *emitted*. We assume the retry attempts are i.i.d. draws from the same candidate-and-gate law with $\Pr[\text{accept}] > 0$, so the emitted unit is a generic candidate conditioned on acceptance. Write $\pi$ for the prevalence of a bad candidate, $a_B = \Pr[\text{accept} \mid \text{bad}]$ for the family's false-accept rate on bad candidates, and $a_G = \Pr[\text{accept} \mid \text{good}]$ for its acceptance rate on good ones (its false-reject rate is $1 - a_G$). An *emitted* unit is one that passed the gate, so its escape probability is conditioned on acceptance: the relevant per-unit quantity is the posterior

$$f \;:=\; \Pr[\text{bad} \mid \text{accepted}] \;=\; \frac{\pi\, a_B}{\pi\, a_B + (1-\pi)\, a_G}. \tag{1}$$

This is exactly Part I's per-segment slip probability $\varepsilon_{\text{seg}}$ (accept-conditioned), read per output unit rather than per checkpoint. Since $f$ is non-decreasing in $a_B$ and a bad artifact in the stealth set is accepted by every gate—hence by any monotone aggregator that accepts on unanimous acceptance ($A(1, \ldots, 1) = 1$), majority included—the family's false-accept rate satisfies $a_B \geq \lambda_{\text{st}}$, so for a *fixed* stack $f$ is floored: with $0 < \pi < 1$ and $a_G \leq 1$,

$$f \;\geq\; \frac{\pi\lambda_{\text{st}}}{\pi\lambda_{\text{st}} + (1-\pi)a_G} \;\geq\; \frac{\pi\lambda_{\text{st}}}{\pi\lambda_{\text{st}} + (1-\pi)} \;>\; 0 \quad \text{whenever } \lambda_{\text{st}} > 0. \tag{2}$$

The $a_G$-dependent middle bound is the sharp floor for a fixed stack; the right-hand ($a_G = 1$) bound is the stack-independent family floor. No amount of same-family verification lowers $a_B$ below $\lambda_{\text{st}}$, so none lowers $f$ below the stack-independent floor. The levers that *do* lower it act on $\lambda_{\text{st}}$ (grounding, diversity, and specifications, §4), on the prevalence $\pi$ (a better executor), or on $a_G$ (fewer false rejects, which dilute the accepted pool).

**Proposition 1** (Series-system escape identity)**.** *Model an output as $n$ emitted units, each independently carrying an undetected fault with fixed probability $f \in [0, 1]$ from (1). Then the probability that the output contains at least one escaped fault is the series-system reliability complement*

$$E(n) = 1 - (1-f)^n \;=\; nf - \binom{n}{2} f^2 + \cdots, \tag{3}$$

*with the union bound $E(n) \leq nf$ (an inequality, tight to first order for $nf \ll 1$).*

*Proof.* Each unit is clean with probability $1 - f$; by independence all $n$ are clean with probability $(1-f)^n$, and $E(n)$ is the exact complement. □

Equation (3) is entirely classical—the reliability of a series system of $n$ independent components [22], equivalently the independent family-wise error of simultaneous inference (the Šidák form [24]) and the same at-least-one-event form as the code model pass@$k = 1-(1-p)^k$ [31, 32] (with escapes in place of successes). It is also, read token-by-token, the much-discussed autoregressive-divergence argument that $\Pr[\text{correct}] = (1-e)^n \to 0$; a recent critique [36] names this exact model "the prevailing assumption" and argues its uniform-independence premise fails because errors concentrate at a sparse set of key tokens. We make no novelty claim for (3); the observation we add is only its reading as the size-axis form of Part I's horizon ceiling.

**Corollary 1** (Zero-escape needs a zero floor)**.** *For* fixed $f$*,* $E(n) \to 0$ *as* $n \to \infty$ *iff* $f = 0$*; for any fixed* $f > 0$*,* $E(n) \to 1$*. (If* $f = f_n$ *varies with* $n$*, the governing condition is* $nf_n \to 0$*.)*

The corollary is the honest ceiling on "no faults at scale." Because $f$ increases in $a_B$ and same-family gates keep $a_B \geq \lambda_{\text{st}}$, once $a_B$ has saturated at the shared blind spot, adding more gates of the *same family* cannot lower $f$ past the floor (2) (Part I [2]). Diversity lowers the floor only through the *reducible* part of the stealth mass: $k$ independent families drive the stealth-limited floor down along $\lambda_{\text{st}}^{(1:k)} = \lambda_{\text{irr}} + (1-\lambda_{\text{irr}})\rho^k \to \lambda_{\text{irr}}$ (Part I's decomposition, with the reducible residuals independent across families conditional on the common irreducible set—distinct from the unconditional exogenous-family product $\mu(\Sigma_F \cap \Sigma_N) = \lambda_F \lambda_N$ of [1]), so the floor on $a_B$—and hence on $f$—approaches its $\lambda_{\text{irr}}$-level value, *not* zero. (Whether the operational $a_B$ itself attains that floor depends on the aggregator; for the unanimous model $a_B = \lambda_{\text{st}}$.) The irreducible floor $\lambda_{\text{irr}}$ yields only to better specifications or reality-grounding; the prevalence $\pi$ yields only to a better executor. Below saturation, same-family depth does reduce the reducible residual—the claim is only that it cannot pass the floor.

**Unification.** Equation (3) is one quantity into which separate results collapse. Because $f$ (1) is precisely Part I's per-segment slip probability $\varepsilon_{\text{seg}}$, the horizon ceiling of Part I is the time-ordered instance of the same geometric law: an output is a sequence of $n = T/s$ segments, each emitted with slip probability $f = \varepsilon_{\text{seg}}$, and end-to-end success $(1-f)^n$ is (3) read along the time axis. The two are the same identity on two different indices, and $\lambda_{\text{st}}$ supplies the floor on $f$ in both. A size-axis generalization to outputs whose units share provenance is the subject of §6; a generalization to arbitrary provenance *graphs* is left open (§8).

**Evidence and caveats.** The identity is illustrated in the companion script `simulate_floor.py` at $f = 0.05$: at $n = 20$ the predicted escape 0.6415 matches the Monte-Carlo value 0.6407 (seed 20260718) within sampling error, and the fit holds across $n \in \{1, 5, 10, 20, 50\}$. Two modelling assumptions are worth naming. First, (3) assumes the $n$ units are *independent*; units passing a common terminal gate share that gate's blind spot, a correlation channel distinct from the provenance sharing of §6—and, *to the extent that channel fits the monotone independent-primitive model of Assumption 3*, it pushes $E$ below the independent value. Second, "output unit" has a clean meaning for structured outputs (fields, claims, functions), but its right definition for continuous or unstructured outputs is open, and $E(n)$ should be read with that caveat where units are not crisply individuated.

## 4 The stealth-mass floor and its epistemic reading

The floor these results rest on is not established here; it is the subject of two companion papers. Part I [2] gives the statistical floor: a gate family's *stealth mass* $\lambda_{\text{st}}$—the fraction of bad artifacts every gate in the family accepts—bounds majority accuracy, and decomposes as $\lambda_{\text{st}} = \lambda_{\text{red}} + \lambda_{\text{irr}}$ into a diversity-reducible part and a specification-bound irreducible floor. The companion paper [1] gives the mechanism: modelling a family $F$ by its representation relation $\sim_F$ (indistinguishability under every prompt), the stealth set $\Sigma_F = \{\text{bad } c : \ c \sim_F c' \text{ for some good } c'\}$—with $\mu$ the fault-generating distribution and $\lambda_F = \mu(\Sigma_F)$—is accepted by every *sound* gate built within $F$, and with nonzero false rejects up to a vanishing margin (its GG1). Consequently within-family generation cannot drive the escape floor below $\lambda_F$—however deep the generation, however many gates check gates, however diverse the prompts (its GG2; we use only this lower-bound direction, which is what our conclusions need, and not GG2's achievability, which requires sufficient within-family coverage). Only two moves lower the floor: importing an *exogenous* family, whose blind spot, when independent, multiplies it ($\mu(\Sigma_F \cap \Sigma_N) = \lambda_F\lambda_N$, driving $\lambda_{\text{red}}$ down), and a *reality-grounded* gate—one deciding by ground truth rather than any family's representation—which removes the grounded part of $\lambda_{\text{irr}}$. We take this floor and its two levers as given.

**An epistemic/diagnosability reading.** The one observation we add is that this floor is a familiar object across several established literatures—worth recording because it locates the phenomenon and supplies external tools. The relation $\sim_F$ is an indistinguishability relation in the possible-worlds sense; the accept set $\{w : [w]_F \cap G \neq \emptyset\}$ (the bad worlds $\sim_F$-linked to some good one, whose $B$-mass is $\lambda_F$) is exactly the rough-set upper approximation of the good set [26]; "a fault is undetectable iff observationally indistinguishable from acceptable behaviour, and duplicating an observation channel without new information cannot restore detection" is the substance of diagnosability and decentralized fault-diagnosis for discrete-event systems [27, 28, 29]; and the pooled vision of diverse families is the *distributed knowledge* of multi-agent epistemic logic [13, 25] (the intersection $\bigcap_i \sim_i$ of the family relations, strictly finer than any one). On the language-model side the same impossibility has just appeared: Mason and Anand [39] prove that a text-only supervisor cannot distinguish grounded from fabricated outputs and that no amount of same-channel supervision resolves it—GG2's within-family invariance in a different formalism—consistent with the empirical "models think alike" correlation-floor findings of [37, 38]. Every one of these readings names the same lever: a distinction must be *imported*—an exogenous family, or reality-grounding—never bred within a family. This is the semantic content behind the assurance framing of §7 ("who ran the check, on whose object" is the informal name for "which family's $\sim_F$ was applied").

## 5 A conditional cost comparison: gating versus scaling

Why gate an unreliable executor rather than wait for, or pay for, a more reliable one? This section gives a model calculation—not a general theorem—comparing the cost of the two routes. It formalizes a claim now argued empirically across the test-time-compute literature [33, 34, 35]: that adding verification is a cheaper lever than scaling the generator. Everything here is conditional on the two cost models we state as assumptions, and neither ingredient is individually new—logarithmic redundancy for a reliability target is von Neumann's [3] and Part I's, and the other side is inverting an empirical power law.

We work throughout in the per-emitted-unit escape probability $f$ of §3, and compare the two routes *for a fixed baseline executor*—so the prevalence $\pi$ is held fixed and the irreducible floor is a constant. In the epistemic model that floor is the posterior value $q_{\text{irr}} = \pi\lambda_{\text{irr}}/(\pi\lambda_{\text{irr}}+(1-\pi)a_G) > 0$, but we take $q_{\text{irr}}$ as an exogenous constant of a fixed specification and executor (on the scaling route, a genuinely better executor lowers $\pi$ and hence $f$ separately—a different lever, out of scope here). We compare the cost of reaching a target a distance

$$\delta \ := \ q - q_{\text{irr}} \ > \ 0$$

above the floor, where $q$ is the required per-unit escape; for an output of $N$ independent units a target output-escape $\varepsilon$ sets $q \approx \varepsilon/N$ (§3). We compare as $\delta \to 0^+$.

**Assumption 1** (Scaling cost, floor-aware)**.** *Driving the executor's own per-unit escape down by compute obeys a power law above the floor, $e(C) = q_{\text{irr}} + \kappa\, C^{-p}$ for constants $\kappa > 0$, $p \in (0, 1]$. We take the exponent from language-model* loss *scaling [14, 15] ($p \approx 0.05$–$0.5$) as a* hypothesis only*: those laws fit cross-entropy loss, not task-error or intent-violation rate, and the loss-to-error map is an extra unstated step.*

**Assumption 2** (Gating cost, floor-aware)**.** *Diverse gate families reduce the* reducible *conditional false-accept rate geometrically: $a_B(n) = \lambda_{\text{irr}} + (a_{B,0} - \lambda_{\text{irr}})\, c^n$ for a per-family multiplier $c \in (0, 1)$ (Part I's decomposition with conditional independence across families; §3). We further assume the good-acceptance rate is bounded away from zero, $a_G(n) \geq a_G^{\min} > 0$, and $0 < \pi < 1$, $f_0 > q_{\text{irr}}$. Under these, the posterior map $a_B \mapsto f$ (1)—a Möbius transform, smooth and increasing with derivative bounded away from $0$ and $\infty$ on the range—carries the geometric decay to the induced escape at the* same rate*,*

$$f_{\text{gate}}(n) - q_{\text{irr}} \;=\; \Theta(c^n),$$

*which is all Proposition 2 uses. We write $f_{\text{gate}}(n) = q_{\text{irr}} + (f_0 - q_{\text{irr}})c^n$ below only as a representative closed form with this rate, not an exact identity; each family costs a fixed $g_F$. The $a_G(n) \geq a_G^{\min}$ hypothesis is load-bearing: if instead conjoining families drives $a_G(n) = \prod_i a_{G,i} \to 0$ (pure unanimous stacking), then for $\lambda_{\text{irr}} > 0$ the posterior $f \to 1$, not $q_{\text{irr}}$—the comparison then fails, and holding $a_G$ up requires a calibrated aggregator, not raw conjunction (see "where the argument stops").*

*Scale.* Solving $e(C) = q_{\text{irr}} + \delta$ under Assumption 1 gives the compute to make the executor itself that reliable,

$$C(\delta) = (\kappa/\delta)^{1/p}, \tag{4}$$

polynomial in $1/\delta$ with a large exponent $1/p \approx 2$–$20$.

*Gate.* Solving $f_{\text{gate}}(n) \leq q_{\text{irr}} + \delta$ gives the number of families,

$$n_{\text{gate}}(\delta) = \max\Big\{0,\ \Big\lceil \tfrac{\ln((f_0 - q_{\text{irr}})/\delta)}{\ln(1/c)} \Big\rceil\Big\} = O\big(\log(1/\delta)\big), \tag{5}$$

logarithmic in $1/\delta$, at total cost $g_F\, n_{\text{gate}}(\delta)$.

**Proposition 2** (Cost separation)**.** *Under Assumptions 1–2, as the target approaches the shared floor ($\delta \to 0^+$) the gating cost is logarithmic and the scaling cost polynomial in $1/\delta$, so their ratio vanishes:*

$$\frac{g_F\, n_{\text{gate}}(\delta)}{C(\delta)} = \frac{g_F\, O(\log(1/\delta))}{(\kappa/\delta)^{1/p}} \;\longrightarrow\; 0.$$

*Proof.* Put $x = 1/\delta \to \infty$. Then $C = \kappa^{1/p} x^{1/p}$ and $g_F\, n_{\text{gate}} \leq g_F(1 + \ln((f_0 - q_{\text{irr}})x)/\ln(1/c))$, so the ratio is $O(\log x)/x^{1/p} \to 0$, since $\ln x = o(x^{1/p})$ for every $p > 0$. □

The separation is meaningful for $\delta \in (0,\ f_0 - q_{\text{irr}}]$; it says nothing at or below the floor, where $\delta \leq 0$ and neither route helps. Table 1 makes the gap concrete for one illustrative choice of constants ($p = 0.1$; $\kappa = 0.1$; $f_0 - q_{\text{irr}} = 0.1$; $c = \frac{1}{2}$; $g_F = 1$); the asymptotic separation, not the row values, is the result, and every entry is reproducible from (4)–(5).

| excess target $\delta = q - q_{\mathrm{irr}}$ | scale compute $C(\delta)$ | gate families $n_{\mathrm{gate}}(\delta)$ |
|---|---|---|
| $10^{-2}$ | $10^{10}$ | 4 |
| $10^{-4}$ | $10^{30}$ | 10 |
| $10^{-6}$ | $10^{50}$ | 17 |
| $10^{-8}$ | $10^{70}$ | 24 |

Table 1: Illustrative cost of reaching an excess target $\delta$ above the shared floor by scaling the executor ($C = (\kappa/\delta)^{1/p}$) versus by gating ($n_{\mathrm{gate}} = \lceil \log_c(\delta/(f_0 - q_{\mathrm{irr}})) \rceil$), for $p = 0.1$, $\kappa = 0.1$, $f_0 - q_{\mathrm{irr}} = 0.1$, $c = \frac{1}{2}$. Scaling compute grows by ten orders of magnitude per decade of $\delta$; gating grows by about three families.

**Where the argument stops, and what it rests on.** This is a *toy asymptotic specialization*, not an operational cost model, and it instantiates the classical redundancy-versus-component-improvement trade-off of reliability engineering [47] in the Part I floor metric; we do not claim the trade-off itself, only its statement here. Four assumptions are load-bearing. First, the power law of Assumption 1 is a hypothesis carried over from loss scaling, not a proven law for task error. Second, $q_{\mathrm{irr}}$ is assumed exogenous and common to both routes; on the scaling route, lowering the executor's error changes the prevalence $\pi$, so a fixed floor there is itself an assumption (Corollary 1's "only specs/grounding lower the floor" holds at a fixed executor; a better executor lowers $\pi$, hence $f$, separately). Third, the comparison counts only executor compute $C$ against gate cost $g_F n$; it omits the *retry cost* of gating—conjoining $n$ families multiplies the good-acceptance rate to $\prod_i a_{G,i}$, so the expected attempts per emitted unit, $1/\Pr[\text{accept}]$, can grow with $n$ and erode the advantage unless false rejects are controlled (e.g. by a majority rather than unanimous aggregator, which then breaks the clean product law)—as well as baseline executor compute, diversity-acquisition cost, and latency. Fourth, Assumption 2 presumes an *unbounded supply* of independent families, exactly the scarce resource the rest of this theory names as binding. Finally, we mention but do *not* rely on the automata metaphor sometimes attached to this comparison: determinizing a non-deterministic automaton by subset construction blows up exponentially [16], and the blow-up is genuinely necessary [30]. That is a statement about representational state complexity, not about paying compute to lower a stochastic error rate; it is at most an illustrative analogy and is not evidence for Proposition 2.

## 6 Correlated escape under shared provenance

The escape identity (§3) assumes independent output units. Real outputs often share upstream provenance: many fields derived from one computation, many claims resting on one retrieved fact. A shared upstream fault is then a *single* event, not $n$ independent ones, and one expects correlated escape to sit *below* the independent bound. This is a textbook consequence of the classical theory of *association* of random variables [23] and of coherent-system reliability [22]: monotone functions of independent variables are associated, and associated components satisfy exactly the series-system inequality below. We do not claim the inequality; our contribution is only its application to output provenance and the packaged strictness characterization, which the companion papers state only for the independent-unit case.

**Assumption 3** (Monotone provenance model). *The escape of unit $j$ is an indicator $I_j = \phi_j(X) \in \{0,1\}$, where $X = (X_1, \ldots, X_M) \in \{0,1\}^M$ is a vector of* mutually independent *primitive fault/leak variables—the faults of the shared upstream parents together with each unit's own local fault source, each surviving its gates independently—and each $\phi_j$ is coordinatewise non-decreasing. Monotonicity* *encodes that a primitive fault can only ever cause, never prevent, an escape: $I_j = 1$ if any parent feeding unit $j$ leaks past its upstream gates, or its local source does. Write $f_j = \Pr[I_j = 1]$, and $f$ for the common value when the units are exchangeable.*

The bound is the association / series-system inequality [23, 22]; we state the product form used here and prove it for completeness (its cleanest source is [23]).

**Lemma 1** (Association product form; Esary–Proschan–Walkup)**.** *Let $X = (X_1, \ldots, X_M)$ have independent coordinates, and let $g_1, \ldots, g_n : \{0,1\}^M \to [0, \infty)$ be bounded and all coordinatewise non-increasing (equivalently, by symmetry, all non-decreasing). Then $\mathbb{E}\big[\prod_{j=1}^n g_j(X)\big] \geq \prod_{j=1}^n \mathbb{E}[g_j(X)]$.*

*Proof.* The base case $n = 2$ is Harris's inequality [19] (a special case of FKG [20]): for independent coordinates, two bounded functions monotone in the same direction satisfy $\mathbb{E}[g_1 g_2] \geq \mathbb{E}[g_1]\mathbb{E}[g_2]$. The non-decreasing and non-increasing cases are interchanged by replacing each $X_i$ with $1 - X_i$, which preserves independence and flips every $g_j$ simultaneously; we argue the non-increasing case. For the induction, the partial product $G = \prod_{j<n} g_j$ is again non-negative, bounded, and non-increasing (a product of non-negative non-increasing functions), so the $n = 2$ case applied to $(G, g_n)$ and the induction hypothesis give $\mathbb{E}[\prod_{j \leq n} g_j] = \mathbb{E}[G\, g_n] \geq \mathbb{E}[G]\,\mathbb{E}[g_n] \geq \big(\prod_{j<n} \mathbb{E}[g_j]\big)\mathbb{E}[g_n]$. This is the closure of association under coordinatewise-monotone maps and the series-system bound of [23]. □

**Proposition 3** (Correlated-escape bound)**.** *Under Assumption 3, for* any *number of shared parents,*

$$E_{\mathrm{corr}}(n) \;=\; 1 - \Pr\Big[\textstyle\bigcap_{j=1}^n \{I_j = 0\}\Big] \;\leq\; 1 - \textstyle\prod_{j=1}^n (1 - f_j), \tag{6}$$

*which equals $1 - (1 - f)^n$ when the units share a common marginal $f$. Assume in addition every unit has $f_j < 1$. Then the bound is strict as soon as some shared parent $X_s$ is* jointly pivotal *for two distinct units $j, k$—i.e. $X_s$ is non-degenerate and, on a positive-probability set of the other coordinates $X_{-s}$, both $\phi_j$ and $\phi_k$ are non-constant in $X_s$—and $n \geq 2$. Under the same $f_j < 1$ nondegeneracy, equality holds* iff *the $I_j$ are mutually independent.*

*Proof.* Each $g_j := 1 - I_j = 1 - \phi_j(X)$ is a bounded, non-negative, coordinatewise non-increasing function of the independent vector $X$. Lemma 1 gives

$$\Pr\Big[\textstyle\bigcap_j \{I_j = 0\}\Big] = \mathbb{E}\Big[\textstyle\prod_j (1 - I_j)\Big] \;\geq\; \textstyle\prod_j \mathbb{E}[1 - I_j] = \prod_j (1 - f_j),$$

and taking complements yields (6); with exchangeable units $\prod_j (1 - f_j) = (1 - f)^n$. This uses only monotonicity and independence of the *primitives*, so it holds for any number of shared parents.

*Strictness.* Let $U = 1 - I_j$, $V = 1 - I_k$. By the law of total covariance,

$$\mathrm{Cov}(U, V) = \underbrace{\mathbb{E}\big[\,\mathrm{Cov}(U, V \mid X_{-s})\big]}_{\text{(I)}} + \underbrace{\mathrm{Cov}\left(\mathbb{E}[U \mid X_{-s}],\; \mathbb{E}[V \mid X_{-s}]\right)}_{\text{(II)}}.$$

Term (II) $\geq 0$: $\mathbb{E}[U \mid X_{-s}]$ and $\mathbb{E}[V \mid X_{-s}]$ are non-increasing functions of the independent coordinates $X_{-s}$, so Harris applies. Term (I) $> 0$: under joint pivotality, on a positive-probability set of $X_{-s}$ both $U$ and $V$ are non-constant and same-direction monotone in the non-degenerate $X_s$, giving strictly positive conditional covariance there and $\geq 0$ elsewhere. Hence $\mathrm{Cov}(U, V) > 0$, i.e. $\mathbb{E}[UV] > (1 - f_j)(1 - f_k)$. Treating $h := UV$ as one non-negative non-increasing factor and applying Lemma 1 to $\{h\} \cup \{1 - I_l : l \neq j, k\}$,

$$\mathbb{E}\Big[\textstyle\prod_l (1 - I_l)\Big] \;\geq\; \mathbb{E}[h] \textstyle\prod_{l \neq j,k} \mathbb{E}[1 - I_l] \;>\; (1 - f_j)(1 - f_k) \textstyle\prod_{l \neq j,k} (1 - f_l) = \prod_l (1 - f_l),$$

where the strict step uses $f_l < 1$ so that $\prod_{l \neq j,k}(1 - f_l) > 0$. Thus ([6]) is strict. *Equality:* mutual independence gives $\mathbb{E}[\prod_j (1 - I_j)] = \prod_j (1 - f_j)$ at once. Conversely, suppose equality holds and all $f_l < 1$. For *any* subset $S \subseteq \{1, \dots, n\}$, group $h_S = \prod_{j \in S}(1 - I_j)$ (non-negative and non-increasing) and apply Lemma 1 to $\{h_S\} \cup \{1 - I_l : l \notin S\}$:

$$\prod_l (1 - f_l) = \mathbb{E}\Big[\prod_l (1 - I_l)\Big] \;\geq\; \mathbb{E}[h_S] \prod_{l \notin S}(1 - f_l) \;\geq\; \prod_l (1 - f_l),$$

so both inequalities are equalities; since $\prod_{l \notin S}(1 - f_l) > 0$ (all $f_l < 1$), this forces $\mathbb{E}[h_S] = \prod_{j \in S}(1 - f_j)$, i.e. $\Pr[\bigcap_{j \in S}\{I_j = 0\}] = \prod_{j \in S}(1 - f_j)$. Holding for every $S$, this is mutual independence of the events $\{I_j = 0\}$, hence of the $I_j$. (We do not need, or invoke, any pairwise-implies-mutual result for associated variables; the subset form of Lemma 1 delivers mutual independence directly.) □

*Remark* 1 (The multi-parent case). One might ask whether "strictly below independent" survives when a unit has $k \geq 3$ shared parents, where multi-parent dependence might turn an escape correlation negative. It cannot: the escape indicators are monotone functions of *independent* primitives, so association holds for *any* $k$ [23], and strictness is governed only by joint pivotality (with $f_l < 1$), not by parent count. This was never open in the classical association literature—EPW places no restriction on the number of shared inputs; we simply make it explicit here for the provenance setting.

*Remark* 2 (Interpretation). Equation (6) compares dependence structures *at fixed unit marginals*: positive common-cause dependence *clusters* escapes, so the probability of at least one escape is no larger than under independence. It does *not* say shared provenance "helps reliability." Shared causes raise the chance of simultaneous multi-unit failure, destroy redundancy benefits, and in practice often *raise* the marginals themselves—the harmful regime studied under common-cause failure. The bound is a statement about the OR-event under fixed marginals, nothing more.

**Evidence and scope.** The bound is illustrated in `simulate_floor.py` on a one-shared-parent event model (a shared upstream fault present with $q_{\text{up}} = 0.3$, surviving the upstream gate with $a_{\text{up}} = 0.247$ and each output's sink with $a_{\text{sink}} = 0.247$; independent local faults with $q_{\text{loc}} = 0.05$, per-unit marginal $f_{\text{marg}} = 0.0304$). At $n = 20$ the correlated escape is 0.2777 (exact; Monte-Carlo 0.2785, seed 20260718) against the independent bound 0.4610—a gap of 0.183 that widens across the displayed range (both probabilities tend to 1 as $n \to \infty$, so the gap eventually narrows). The *problem* of where to place verification on a provenance graph is already public [18]; we stop at the bound and leave any allocation policy out of scope (§8).

# 7 Grading assurance: a framing the floor explains

The results here rest on one object—an escape floor that internal effort cannot cross—and that floor has a practical corollary for how one should *talk* about whether a gated system "works." We propose, not prove, an organizing frame with two parts, both deliberately modest and both already anticipated in the assurance literature; what this paper adds is the observation that the floor results explain *why* each holds.

## 7.1 A proposed evidence ladder

Claims that a built system "works" or is "better than before" can be graded on an ordinal ladder by *who ran the check, on whose object, for whose benefit*: (R1) **built and internally verified**—demonstrated by a re-runnable check the builder wrote, on an object the builder controls;

(R2) **independently reproduced**—a second party reproduces the result from the builder's artifact without the builder's help; (R3) **externally validated**—exercised by a party outside the builder's control, on a workload the builder did not author; (R4) **value-validated**—that external party derives measured value from it, not merely confirms it runs. We offer this as a *proposed* taxonomy, not an exhaustive law, and it is a coarsening of frameworks that grade along related independence and maturity axes: independent verification and validation (IV&V, graded by evaluator independence), the reproduced/replicated distinction of artifact-evaluation badging (graded by who ran the check, on whose artifacts), technology-readiness levels [40] (maturity and operating environment), typed assurance-case evidence [41], evidence-based levels of evidence, standardized AI assurance levels [46], and the evidence-licensing framework of Li [43]. We claim no new taxonomy; the axes—evaluator independence, workload independence, and demonstrated utility—are in fact partly orthogonal, so the rungs should be read as a rough, claim-relative progression rather than a total order (a weak R4 signal is not automatically stronger than a rigorous R1 proof). Its only virtue is a cheap test—the three questions above—for the common overclaim of quoting internal rigor as external assurance.

The floor results say why the rungs cannot collapse into internal rigor, though the link is a heuristic proxy, not a theorem. A system whose internal checks all share one blind structure can have arbitrarily many of them and still sit at R1: same-family depth cannot lower the escape floor (§3, Cor. 1; the self-generation floor of [1]). Moving outward is an *opportunity* to add a genuinely new indistinguishability relation—an external validator (R3) or paying user (R4) exercises the system under assumptions and workloads the builder's gates never saw—but organizational independence does not *guarantee* epistemic non-redundancy: an R2 reproducer re-running the builder's own check largely re-applies the builder's relation and mainly hardens the *evidence* (against fraud or setup error), and an external party can share the builder's assumptions and add nothing. Actual floor reduction requires an imported family $N$ whose stealth set $\Sigma_N$ genuinely excludes part of the existing blind spot, $\mu(\Sigma_F \setminus \Sigma_N) > 0$ (§4); the ladder is a proxy for seeking one, not a proof that a higher rung supplies it.

### 7.2 Typed evidence should not collapse to a scalar

A pipeline that mixes a conditional formal proof, a deterministic test or compile result, a probabilistic model-judgment (an LLM or human reviewer's opinion), and a provenance claim should not report a single external-facing scalar—a probability or "trust score"—folding all four together. This is standard in safety-case practice, where confidence is argued from typed, unmerged evidence rather than reduced to one attribute [41, 45], and it is being restated for language-model pipelines as typed evidence graphs [44]. (Evidence-combination formalisms such as Dempster–Shafer [42] do produce numbers, but they retain ignorance and conflict rather than collapsing to a single score—an illustration that a defensible aggregate must keep its semantics, not that no number may ever appear.) The rule is thus not "never quote a number" but: any external aggregate must carry explicit semantics and stay linked to the typed evidence, provenance, and assumptions it came from; a bare scalar that hides evidence type is what must not be surfaced.

The epistemic model of §4 offers a useful interpretation. Different evidence types are different indistinguishability relations with different blind spots; a *bare* scalar fold is a projection that does not ordinarily expose which part of the confidence is load-bearing under which conditions—the structure the distributed-knowledge reading of §4 retains. A conditional theorem and a measured reviewer rate are sound under different envelopes; averaging them into an unlabelled number reports neither honestly. Non-collapse is the reporting-side image of the paper's recurring fact that diversity lives in the *structure* of what different gates can and cannot see, not in a scalar. Neither the ladder nor this rule is new—each is a light repackaging of assurance-case and evidence-theory practice [40, 41, 42]; the contribution is only the link to the floor.

## 8 Open problems

Five questions are adjacent to the results above but not settled here; we state them as open, with no result claimed, to mark the intended shape of the theory.

1. **A provenance-graph escape law.** The escape identity (§3) treats independent units and §6 treats units that are monotone Boolean functions of arbitrary *independent* primitive faults (any number of shared parents, one layer). The genuinely open case is a full provenance DAG with *internal* gates, statistically *dependent* primitives, recomputation/retries, and non-monotone repairs, and a sink-floor law relating escape to graph structure and per-node floors. We regard this as a prove-first item and do not include it here.

2. **An exact finite-horizon escape process.** Equations (3) and (6) are static bounds; the exact escape probability of a checkpointed process with recomputation is naturally a finite-horizon Markov decision problem whose optimal gating policy would refine the asymptotics of §5. This is in progress and deferred.

3. **Reviewer–producer capability matching.** The results here treat the executor and its gates as given; how gate capability should be matched to executor capability (when a weaker but diverse reviewer helps, when it does not) is open.

4. **A telemetry estimator for the stealth mass.** Part I notes that the reducible stealth mass is partly estimable from a system's own logs and targeted audits; a principled estimator for $\lambda_{\mathrm{st}}$ from telemetry, with an identifiability analysis in the latent-class tradition of Hui and Walter [21] (estimating error rates without a gold standard), is left to future work. This is the quantity that would make the stealth mass predictable rather than only observable.

5. **Checker strength.** The framing of §7 treats gates as given. A separate question is how the *strength of the checking procedure*—whether the property it checks is decidable now, corroborable soon, or gradeable only later—bounds what "verified" can honestly mean for that check's output, and whether it predicts when a checkable-notation effort finds a consumer. We regard this as a single idea deserving its own careful treatment and do not develop it here.

## 9 Related work

This paper builds on Part I [2] and its companion [1] (the statistical and mechanistic accounts of the floor, respectively), and inherits their lineage: reliable computation from unreliable components [3, 4], the quantum threshold theorem [5], Condorcet's jury theorem [6], the Young–Daly checkpoint interval [7, 8], and the $N$-version-programming tradition on coincident failure [9, 10, 11], whose contemporary agentic echo is the large-scale $N$-version coding-agent study of Ron, Baudry and Monperrus [12].

For the specific results here, we attribute the prior art plainly, since the mathematics in each is classical. §3: the escape identity (3) is series-system reliability [22], the Šidák independent-error form [24], and the at-least-one form of pass@$k$ [31, 32]; token-wise it is the autoregressive-divergence argument whose independence premise [36] disputes. The correlation floor it feeds is actively studied—Turkmen, Buyukates and Bastopcu [17] derive an information-theoretic ensemble error floor under a Gaussian-copula model, and "models think alike" correlation is measured by [37, 38]. §4: the stealth-set floor, its inheritance and self-generation invariance, and its two levers are established in the companion [1] (statistically in Part I); we only record its reading in possible-worlds epistemic logic [13, 25], rough-set upper approximation [26], and discrete-event diagnosability [27, 28, 29], noting the language-model impossibility of Mason–Anand [39]. §5: the comparison formalizes a claim argued empirically across the test-time-compute literature [33, 34, 35], resting on loss scaling [14, 15]; the automaton-determinization blow-up [16, 30] is a metaphor, not evidence. §6: the bound is the classical association / series-system inequality [23, 19, 20, 22]; the provenance-placement *problem* is treated by Sherlock [18]; ours is only the application to output provenance and the packaged strictness characterization.

# 10 Conclusion

The classical tools here converge on one object, the verification floor of Part I and [1]. At fixed prevalence, same-family verification cannot lower a gate family's escape floor (§3; the floor is *set by* Part I's stealth mass, and is not a consequence of the series identity itself, which merely propagates it across output size); a system's own self-generated gates cannot lower the shared blind spot (the self-generation floor of [1]); above that floor, under stated and admittedly strong cost models, gating reaches a target more cheaply than scaling the executor's own error does (Prop. 2); and at fixed marginals, positive provenance dependence only clusters escapes, never raising the chance of at least one relative to independence (Prop. 3). None of these is new mathematics; the contribution is the synthesis—three classical tools trained on the floor of Part I and [1]—plus the one conditional cost calculation. The floor has several levers: diversity lowers the lineage-*reducible* stealth mass, better specifications or reality-grounding lower the irreducible part $\lambda_{\text{irr}}$, and a stronger executor lowers the prevalence $\pi$; "irreducible" is always relative to a fixed executor and specification. Much of this is verification engineering available today, alongside gains in raw model capability.

# Code and data availability

The numerical claims of §§3–6 are reproduced by `simulate_floor.py` at `https://github.com/ivmat/gated-computation-sim`. This paper builds on two companion papers archived on Zenodo: Part I, DOI 10.5281/zenodo.20820968, and the self-generation floor paper [1], DOI 10.5281/zenodo.20837102.

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